No. of Functional Features Included in the Solution

Date	01 October 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features included in the FitBuddy project. Each feature is described with its purpose, implementation status, responsible team member, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component & Owner	Status (Done / In Progress / Pending)	Marks Contribution
1	Personalized User Input	Collects name, user ID, age, weight, fitness goal and workout intensity.	Pavithra M index.html / FastAPI forms	Done	0.55
2	Gemini 7-Day Workout Generation	Generates a structured day-wise workout plan using Gemini 1.5 Pro, including warm-up, exercises, sets/reps and recovery.	Pavithra M gemini_generator.py	Done	0.75
3	Nutrition / Recovery Tip	Provides a concise goal-based nutrition or recovery recommendation using Gemini Flash.	Manisha S gemini_flash_generator.py	Done	0.55
4	Feedback-Based Plan Update	Accepts user feedback such as more cardio or rest days and regenerates the plan using Gemini 1.5 Pro.	Monica G updated_plan.py	Done	0.70
5	User & Plan Database Storage	Stores user details and original/updated AI plans in SQLite using SQLAlchemy.	Monisha Anandan database.py	Done	0.60
6	FastAPI Routing & API Integration	Handles /, /generate-workout, /submit-feedback and /view-all-users routes and connects backend logic with Gemini services.	Monisha Anandan routes.py	Done	0.65
7	Responsive Frontend & Jinja2 Templates	Provides index.html, result.html and all_users.html with dynamic rendering, responsive styling and feedback forms.	Mridula M HTML/CSS/Jinja2	Done	0.65
8	Admin View, Deployment & Testing	Provides admin visibility of users/plans and supports local deployment and verification with Uvicorn and FastAPI docs.	Mridula M all_users.html / deployment	Done	0.55

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2

Features Tested & Verified	8 (local deployment and endpoint/UI verification)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Personalized input form; result page; responsive HTML/CSS; feedback form
2	Backend / Logic	3	FastAPI routing; workout generation; feedback-based plan update
3	Database / Storage	1	SQLite + SQLAlchemy user and plan storage
4	API / Integration	1	Google Gemini 1.5 Pro and Gemini Flash integration
5	Security / Authentication	0	No dedicated authentication feature is described in the project document

Team Work Allocation

Team Member	Assigned Work
Pavithra M	Gemini model selection, application architecture, user input flow, and personalized 7-day workout generation.
Manisha S	Nutrition/recovery tip generation using Gemini Flash and supporting core AI functionality.
Monica G	Feedback-based workout plan updating using Gemini 1.5 Pro and related update logic.
Monisha Anandan	SQLite/SQLAlchemy data storage and FastAPI routes/API integration.
Mridula M	Frontend UI, Jinja2 templates, admin view, local deployment and testing/verification.

Project basis: The feature allocation and descriptions are derived from the supplied FitBuddy project document, including its Gemini integration, FastAPI backend, SQLite/SQLAlchemy storage, Jinja2 frontend, admin view, feedback update flow, and local deployment activities.